CHAPTER 4

RESULT AND FINDINGS

This chapter explains the results and findings of the Centralized Web-Based System for Stress Induction Methods. This chapter cover all three phases of the project which are requirements gathering and analysis, system design, and project implementation process which focusing on the achieved and deliverables and to fulfil the objectives of the project.

4.1 Findings Requirements and Analysis Phase

In this phase, the primary activity is to gather all requirements for this project. Requirement was collected from Mr. Muhammad Hamiz bin Mohd Radzi, the researcher for stress research, help to understand the current research methodology, identify pain points, and determine the system features to ensure that the system follow the researcher requirement.

4.1.1 Conducting an Interview

To get all requirements accurately, a two-phase interview approach was performed. Phase 1 were focusing on the current problem and research methodology, while Phase 2 focused on technical requirements for the system. During the interview session, several questions were asked and recorded. Table 4.1 shows the results of this interview session.

Table 4.1 The analysis from the interview session with Mr. Hamiz

	Question	Answer
1.	How are this stress tests currently implemented?	We use E-Prime for Stroop, a separate PowerPoint for TSST instructions, and a physical stopwatch for timing.
2.	How do you currently record participant responses and reaction times?	Stroop data is saved in a text file, but TSST arithmetic errors are recorded manually on paper by an observer.
3.	Should researchers be able to create multiple experiment templates,	Yes, templates are essential for standardization and repeating studies

	modify task parameters (time limit, difficulty, stimuli type), and save configurations for reuse?	
4.	What user roles should the system support (e.g., Admin, Researcher, Participant), and what permissions should each have?	Admin manages users and study settings; Researchers configure studies and export data; Participants only take tests.
5.	Do you need digital informed consent and debriefing pages within the system?	We also need to store consent timestamp and version of the consent form.

Based on the interview session, Mr. Hamiz explained that the traditional approach required multiple change in platform and manual timing tools disrupt the experience flow and compromises data integrity. Furthermore, manual data transcription and the lack of synchronized video recording can create a human error and inefficiencies for researchers.

4.1.2 Analysed Requirements from Interview Session

All data was compiled for analysis and to create diagram based on the information obtained from the interview sessions. This included the System Sequence Diagram (SSD), Domain Class Diagram (DCD), and Use Case Diagram (UCD). The results of this requirements analysis are presented in Table 4.2.

Table 4.2 Requirement Gathering Result

	Answer	Findings	Actor
1.	We use E-Prime for Stroop, a separate PowerPoint for TSST instructions, and a physical stopwatch for timing. Switching programs breaks the flow, allowing	A centralized, guided system that integrates the Stroop Test, MIST, and TSST into a single continuous web session. Use case involved: a) Perform Test	Participant

	participant stress levels to drop.		
2.	Stroop data is saved in a text file, but TSST arithmetic errors are recorded manually on paper by an observer. Transcribing paper notes into Excel often leads to typos.	The system must automatically capture responses, score metrics (reaction time, errors), and securely synchronize timestamps with video feeds. Use case involved: a) View Result b) Generate Report c) Export Test Data	a) Administrator b) Researcher
3.	Yes, templates are essential for standardization and repeating studies.	A template creation module that allows modification of task parameters (e.g., time limits, difficulty, stimuli types) and saves them for reuse. Use case involved: a) Manage Test	a) Administrator b) Researcher
4.	Admin manages users and study settings; Researchers configure studies and export data; Participants only take tests.	A robust role-based access control (RBAC) system with distinct dashboards and permissions for each user type. Use case involved: a) Login b) Create User Account c) View User Account d) Update User Account e) Delete User Account	a) Administrator b) Researcher
5.	We also need to store consent timestamp and version of the consent form.	An onboarding module that records the participant's consent timestamp and the specific version of the consent form used. Use case involved: - Choose Participant -Manage Participant	a) Administrator b) Researcher

Table 4.2 show every operational challenge identified during the interview has a targeted system solution. Three primary actors were identified for the Web-Based System for Stress Induction Method which are an Administrator to exclusively manage user accounts and overall system settings, a Researcher to manage test templates, handle participant records, and export data, and a Participant who accesses the system solely to perform the stress induction tasks.

4.1.3 Use Case Diagram

Based on the analyse requirements, the use case diagram has been created. The figure 4.1 shown the use case diagram for this project.

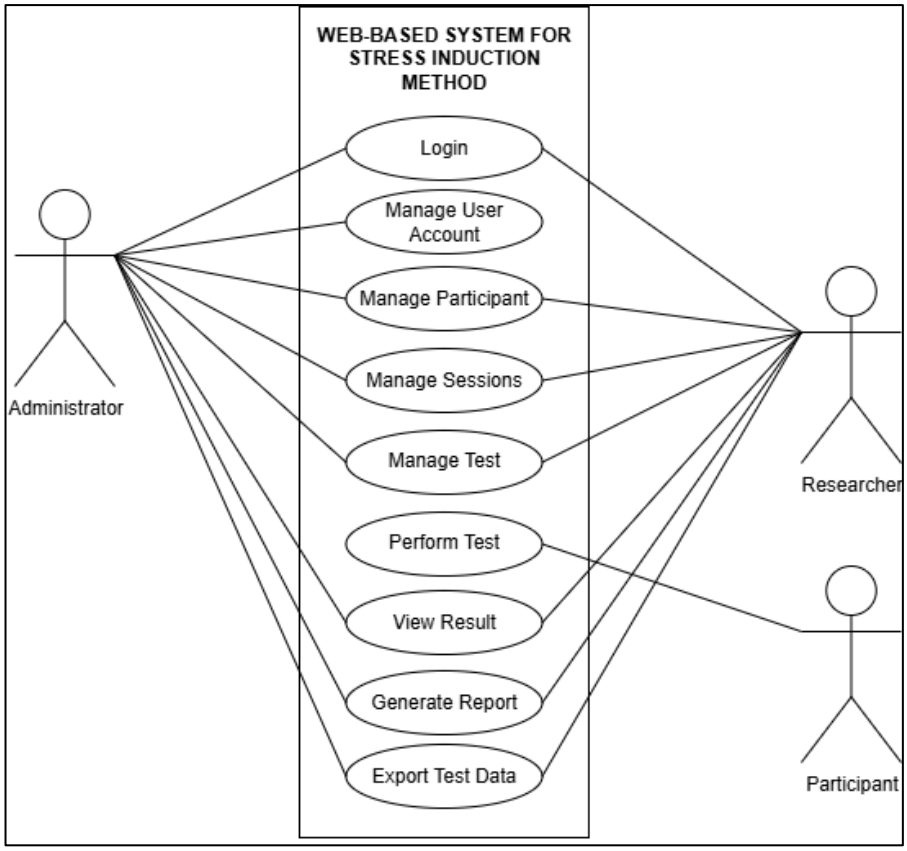


Figure 4.1 Use case diagram for Web-Based System for Stress Induction Methods

The use case diagram in Figure 4.1 shows a total of 9 use cases, with each actor having respective use case functions. Table 4.3 explains the function of each use case.

Table 4.3 Use Case Functionality

Actor	Use Case No.	Use Case	Description
Administrator, Researcher	UCD001	Login	Allows the administrator and researcher to securely authenticate and access the platform's.
Administrator	UCD002	Manage User Account	Enables the administrator to add, view, update, or remove user access.
Administrator, Researcher	UCD003	Manage Participant	Allows for the secure management of participant and details.
Administrator, Researcher	UCD004	Manage Sessions	Enables the management of testing schedules, timetables, and tracking which participants are registered to participate in specific sessions.
Administrator, Researcher	UCD005	Manage Test	Allows the configuration, oversight, and active management of the specific stress induction tests assigned to a participant.
Administrator, Researcher	UCD006	View Result	Enables the instant review of automatically captured metrics, such as reaction times and arithmetic errors, directly within the platform.
Administrator, Researcher	UCD007	Generate Report	Allows the automatic compilation of participant performance into a structured, downloadable summary report.
Administrator, Researcher	UCD008	Export Test Data	Enables the exporting of synchronized raw test data for external statistical analysis.
Participant	UCD009	Perform Test	Allows the participant to undergo the assigned stress induction tasks within a single, continuous web session.

The use cases describe how these actors work with the system to accomplish different goals. Each use case is thoroughly described in the table, which also shows the relationship and procedures that are involved.